

JECRC UNIVERSITY, JAIPUR
Session: Jan-June 2026
Subject: Mathematics-II (DMA025A)
B. Tech - II Semester
Tutorial Sheet– 7
UNIT-4

CO4: To understand complex variables, limits, continuity, differentiability, Cauchy–Riemann equations, analytic and harmonic functions.

Part A ($5 \times 2 = 10$ Marks)

- A1.** Define a complex variable and a complex function. Give one example.
A2. Define the limit of a complex function. How does it differ from the limit of a real function?
A3. Define continuity of a complex function at a point $z = z_0$.
A4. State the Cauchy–Riemann equations in Cartesian form.
A5. Define analytic function. Give one example.

Part B ($3 \times 7 = 21$ Marks)

- B1.** Check whether the function $f(z) = z^2 + 3z + 2$ is analytic and find its derivative.
Ans.: $f'(z) = 2z + 3$
B2. Verify the Cauchy–Riemann equations in Cartesian form for $f(z) = x^2 - y^2 + i(2xy)$ and hence find $\frac{df}{dz}$.
Ans.: $\frac{df}{dz} = 2z$
B3. Show that $u(x, y) = x^2 - y^2$ is a harmonic function and find its harmonic conjugate.
Ans.: $v(x, y) = 2xy$

Part C ($3 \times 11 = 33$ Marks)

- C1.** Find the analytic function whose real part is $u = e^x (x \cos y - y \sin y)$.
Ans.: $f(z) = ze^z$,
C2. If $u(x, y) = x^3 - 3xy^2$, show that u is harmonic and find its harmonic conjugate.
Ans.: $v(x, y) = 3x^2y - y^3$
C3. Prove that if $f(z)$ is analytic, then its real and imaginary parts are harmonic.